TERTIARY INFOTECH ACADEMY

LAB 02
Automate Event-Driven Incident Triage

AWS Certified SysOps Administrator Associate Training
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Scenario

Northstar Retail needs to detect an unapproved security-group change quickly, enrich the event and notify the operator without making an unsafe automatic network change.

Objective

Apply K3, K4, A1 and A4 by routing a controlled AWS event through EventBridge, invoking a least-privilege remediation function and preserving an auditable triage record.

Deliverable

An EventBridge rule, Lambda triage function, test event, CloudWatch Logs evidence and incident decision record.

Services and tools

EventBridge · Lambda · CloudTrail · CloudWatch Logs · IAM

Safety and cost boundary

- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Review the architecture JPEG and `event-pattern.json`; identify event source, API names, principal context and target.

```
python3 -m json.tool event-pattern.json
```

Evidence note: _____

Step 2 — Inspect `triage.py`; confirm it records event identity and never modifies the security group.

```
python3 -m py_compile triage.py
```

Evidence note: _____

Step 3 — Create the trainer-approved execution role from `lambda-role-policy.json`; replace only the account-specific trust and resource values.

```
python3 -m json.tool lambda-role-policy.json
```

Evidence note: _____

Step 4 — Package and create the Lambda function with a unique learner suffix.

```
zip -j triage.zip triage.py
aws lambda create-function --function-name SysOps-Lab02-Triage-REPLACE_SUFFIX --runtime
python3.13 --handler triage.handler --zip-file fileb://triage.zip --role REPLACE_ROLE_ARN
```

Evidence note: _____

Step 5 — Create the EventBridge rule and add the Lambda target; grant EventBridge permission to invoke only this function.

```
aws events put-rule --name SysOps-Lab02-SGChange --event-pattern file://event-pattern.json --state ENABLED
```

Evidence note: _____

Step 6 — Invoke the function with `test-event.json`; save the response and locate the matching CloudWatch Logs event.

```
aws lambda invoke --function-name SysOps-Lab02-Triage-REPLACE_SUFFIX --payload fileb://test-event.json evidence/invoke-response.json
```

Evidence note: _____

Step 7 — Generate one trainer-approved learner-owned security-group change, then verify the real event arrives through EventBridge.

```
aws logs tail /aws/lambda/SysOps-Lab02-Triage-REPLACE_SUFFIX --since 10m | tee evidence/triage-log.txt
```

Evidence note: _____

Step 8 — Complete `TRIAGE-DECISION.md`: impact, evidence, safe next action, authority, verification and escalation.

```
open TRIAGE-DECISION.md
```

Evidence note: _____

Step 9 — Capture EventBridge rule, Lambda monitor and log evidence as JPEG screenshots.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 10 — Run validation; delete the rule, target, permission, function and learner-owned test change.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The test and real events reach the Lambda target, logs contain the event ID and actor context, no automatic network modification occurs, the decision record names authority and verification, and validation emits LAB 02 PASS.

Acceptance criteria

- Event pattern is limited to named API calls
- Lambda role is least privilege
- Function performs triage only
- Event ID and UTC time are preserved
- Cleanup removes all learner-created automation

Evidence checklist

□ EventBridge rule export

□ Lambda invocation response

□ CloudWatch Logs export

□ JPEG screenshots

□ completed TRIAGE-DECISION.md

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.